Our first midterm exam will be held on Friday, February 23rd in class. This exam will cover everything we have learned from Chapters 1 through 5 of our textbook. You will have the entire class period to work the exam. Some of you may finish early, and if so, you are free to leave once your exam has been submitted. Exams will be handwritten.

You may use one double-sided 8.5 x 11in sheet of notes. You may use a calculator, but not your phone. If you need to borrow a calculator, please ask. No collaboration is allowed as this exam should assess your personal understanding so that I can properly award you credit for your learning.

To study for this test, I strongly recommend that you pull out your old homework assignments, labs, and quizzes, hide the answers from yourself and re-do the work, checking your answers against the solutions that are posted on Blackboard. The exercise sections in our textbook are another good source of problems. Remember that answers to odd-numbered exercises are at the "back" of the book.

Things to know and be able to do

Data basics

- Be able to identify the cases and variables in a dataset and identify variables as numerical or categorical.
- Be able to word variables carefully and appropriately for the cases in a dataset.
- Be able to create a hypothetical first few rows of a dataset for a given study in a "tidy data" manner (cases in rows, variables in columns).
- Be able to pose an appropriate research question about one variable or about more than one variable in a dataset.

Study design: Generalization

- Be able to identify the population and the sample in a study.
- Be able to identify parameters and statistics from a study.
- Be able to discuss several different types of random sampling methods. Make sure you are especially able to explain how to select a simple random sample.
- Be able to explain why random sampling is so important.
- Be able to recognize sampling bias and explain why it is necessary to avoid sampling bias.
- Given a particular study, be able to discuss whether the study lends itself to generalization. (We have had several homework and quiz questions along these lines.)

Study design: Causation

- Be able to identify variables as explanatory or response and explain how you can tell.
- Be able to identify a study as an experiment or an observational study and how you can tell.
- Be able to comment carefully on whether a study can be used to establish causal relationships.
- Be able to propose a possible confounding variable, and explain its connection to the explanatory and response variables in a way that gives an alternative explanation for the connection between the explanatory and response variables.
- Be able to identify a study as being blind or double-blind and explain the importance of this.

Study design: Big picture

- Be able to explain how you would design a study from start to finish so that it lends itself to generalization to the population of interest and to establishing causal relationships. (Hint: Surveys are no good here! Why not?)
- Given the description of a study, be able to critically assess its worth, point out potential flaws, and suggest improvements in the design.

Describing numerical variables

- Given a visualization of the distribution of a single numerical variable, be able to write a description of the distribution, first writing a who/what/where/when sentence, and then continuing the paragraph addressing center, shape, spread, outliers, and any other interesting features. All such descriptions should be in context, and should be supported with numbers from the graph or from computed statistics.
- Be able to compare the distributions of a numerical variable grouped by a categorical variable. Comparisons should address the characteristics outlined in the above bullet point.
- Be able to use appropriate measures of center (mean or median) and spread (standard deviation or quantiles: (middle 50% and middle 95%)). Remember that you should use different combinations of these depending on whether your distribution is symmetric or skewed. What are those combinations?
- Be able to explain what a robust (or resistant) statistic is and give examples.

Describing categorical variables

- Given a visualization of a categorical variable, or a pair of categorical variables, be able to write a paragraph highlighting key features of the distribution, speaking in context, and supporting your points with numbers drawn from the graph.
- Given a contingency table, be able to compute row and column proportions to back up claims you make about a study.
- Be able to comment on whether there is an association between two categorical variables, given a visualization or a contingency table for these variables.